

Worksheet

Do the following series converge or diverge? Be sure to state any tests used and show all work needed apply them.

1. $\sum_{k=0}^{\infty} \frac{4}{5k-3}$

2. $\sum_{k=1}^{\infty} \frac{1}{k^2+5}$

3. $\sum_{k=0}^{\infty} \frac{4}{k!}$

4. $\sum_{k=1}^{\infty} \frac{3^{2k+1}}{2^k 7^{k-1}}$

5. $\sum_{k=1}^{\infty} \frac{3^k}{2^k + 5k^2}$

6. $\sum_{k=1}^{\infty} \frac{\sqrt{k}+4}{k^2-2}$

7. $\sum_{k=2}^{\infty} \left(1 + \frac{2}{k}\right)^k$

8. $\sum_{k=1}^{\infty} \frac{3^k}{(2k)^k}$

9. $\sum_{k=2}^{\infty} \frac{3 \ln(k+1)}{\sqrt{k}}$

10. $\sum_{k=1}^{\infty} \frac{3^k}{(2k+1)!}$